

Correspondence

Comments on "Leading-Zero Anticipatory Logic for High-Speed Floating Point Addition"

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I have read with a great interest the above article¹ by H. Suzuki *et al.* I am familiar with their work, and I found their approach interesting. The idea used to simplify the leading zero anticipator (LZA) I found innovative and an improvement over the one used in the IBM RS/6000.

One of the critical parts of their LZA circuit is the leading zero detector (referred as leading zero counter in the text) which by itself contributes 40% of the delay of the total. The structure of the LZ counter is shown in Fig. 4 of the above mentioned article¹ (Fig. 1 here). To devise a hierarchical tree structure for the LZ counter, with a small and regular fan-in/fan-out, is the key in making it fast and amenable for CMOS implementation. Though a regular implementation of the LZA circuit (that does not involve "horizontal" propagation of the signal) is a significant contribution to the speed of the proposed circuit, one cannot ignore the contribution of the innovative LZ detector to the LZA design.

However, I have found the LZ counter circuit shown in the Fig. 4 of the paper similar to the LZ counter circuit that I have published previously in the period from 1992–94. Three of the five papers related to the subject were published outside of the United States [2], [3], [5] and of those three, two were published in Asia [3], [5]. I wonder if the authors were aware of this work, and I would be interested in their comments. In particular, I would like to point to Fig. 1(b) and (c) in [4] (Fig. 2 here).

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¹H. Suzuki, H. Morinaka, H. Makino, Y. Nakase, K. Mashiko, and T. Sumi, *IEEE J. Solid-State Circuits*, vol. 31, pp. 1157–1169, Aug. 1996.

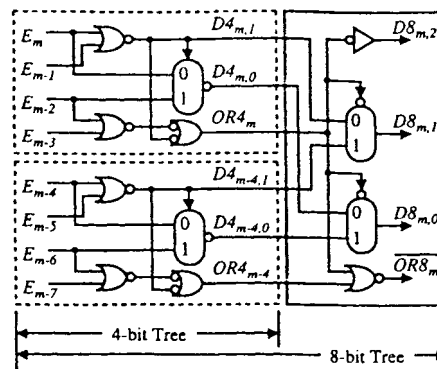


Fig. 1. Basic structure of LZ counter.

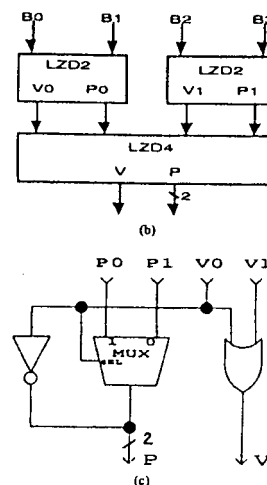


Fig. 2. Design of a 4-b LZD [4].